

Ojasva Mishra

Z Fellow | CMU SCS Researcher | SWE & AI/ML Systems | Quantitative Research Analyst | Robotics & Autonomy
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EDUCATION

University of Florida, Gainesville, FL

Bachelor of Science in Computer Engineering; dual major in Finance

Graduation Date: May 04, 2029

YC Startup School 2026 | **Downingtown STEM Academy, IB Diploma** | **SAT 1590, National Merit Commended**

EXPERIENCE

Drift

November 2025 to Present

AI Platform Engineer (Founding Team)

San Francisco, CA

- Built core infrastructure for an **Antler-backed robotics AI coding workspace used by 1,000+ users** to configure, run, and debug simulations across ROS2, Gazebo, MuJoCo, and Isaac Sim.
- Shipped **World Labs photo-to-3D scene generation**, a license-verified **129-robot catalog**, and autonomous OpenClaw + pexpect evals; traced a production Gazebo freeze to gz-transport UDP multicast.

Carnegie Mellon University, Xu Lab

July 2025 to Present

Research Intern

Pittsburgh, PA

- First author on three robotics and control papers** with Prof. Min Xu and Dr. Xiaolong Wu covering actuator-aware control optimization, adversarial safety filtering, and secure feedback control.
- Validated algorithms on a **Franka Emika Panda**; built multimodal SLAM, path planning, and constrained control experiments spanning perception, planning, and actuation.

Twinly Incorporated

December 2025 to Present

Co-Founder and CEO

San Francisco, CA

- Built Twinly, which models a user's writing, speech, and computer behavior to create **digital clones** that handle work across hundreds of parallel instances; selected for **Z Fellows' 10-builder cohort**, with mentorship from founders of **Netflix, Eventbrite, and Tinder**.
- Shipped production **macOS, Windows, and iOS apps** serving **1,000+ active users, including paying customers**; built the Identity Runtime with app/browser context, persistent memory, and multimodal vision.

AlgoGators Investment Fund

September 2026 to Present

Quantitative Research Analyst

Gainesville, FL

- Selected as **1 of 2 first-year students** for the quantitative research team at an approximately **\$500K AUM** student-run fund; build systematic signals, backtests, and trading strategies.
- Executed a **crack spread strategy with a 64% win rate across live trades**; adjusted positions as hurricane forecasts changed during an active storm.

EV Bots, Inc.

April 2025 to February 2026

AI Robotics Engineer Intern

Chester Springs, PA

- Built the company's **first functional autonomous EV-charging robot prototype** across mechanical design, electronics, perception and control, and closed-loop vehicle connection; designed an autonomous drone-charging dock. Work contributed to multiple provisional patents.

PteroLabs

July 2025 to January 2026

Engineering Intern (UAV Simulation and Autonomy)

Remote, United States

- Built a **mission-grade Unreal Engine 5 UAV simulator** with PX4, ArduPilot, and Betaflight SITL, 6-DOF flight dynamics, FPV, ML mission planning, and object detection; used by the **University of Pennsylvania** for active research.

iPipeline

June 2025 to August 2025

Software Development Intern

Exton, PA

- Shipped **enterprise software infrastructure**: automated test frameworks, AI agents, and developer tooling adopted across engineering teams for validation and regression workflows.

RESEARCH AND PUBLICATIONS

- First author:** [Constraint-Aware Discrete-Time PID Gain Optimization for Robotic Joint Control Under Actuator Saturation](#). **IEEE T-RO**.
- First author:** [Bounding Detector-Accepted Feedback Authority in Torque-Limited Robot Control](#). **IEEE ICRA 2027**.
- First author:** [SCIG++: Risk-Limiting Conformal Safety Filtering for Attack-Resilient Robotic Joint Control Under Saturation](#). **IEEE/RSJ IROS 2026 (video)**.
- [Introduction to Robotics and Programming](#), a 120-page open-access textbook published through FusionBots and indexed on SSRN.

PROJECTS, COMPETITIONS AND LEADERSHIP

- American Rocketry Club, Founder and President:** grew to 60+ members; led payload CAD, propulsion analysis, systems integration, and launches.
- VEX Robotics, 3x World Championship Competitor:** captain, lead programmer, and driver; seven seasons of C++ and Python autonomy, PID tuning, CAD, and sensors.
- Technology Student Association, 2x National Finalist:** 8 national events; captain for 6. Projects: [SAFMS AI/IoT](#), [drone challenge](#), [engineering design](#), [systems control](#), [robotics](#), [VR visualization](#), [video game design](#).
- Hackathons and builds:** YC Transpose, **1st (HIVE)**; Startup in a Weekend Philly, **2nd (GlassBoxAI)**; NexHacks ([Kinetix](#)); [Eyesight](#); [PreventaAI](#); [QuickCitePro](#); [FusionBots](#).
- Other leadership:** Co-founder of [FusionBots](#), a robotics nonprofit serving 200+ students; CTO of [Youth Publications and Socioeconomic Forum](#), leading AI literacy programs across 15+ countries.

TECHNICAL SKILLS

Languages and software: Python, C++, Swift, C#, JavaScript, TypeScript, SQL, Linux, Git, gRPC, React, Next.js, Tauri, Electron

AI and ML: PyTorch, LLM agents, multimodal systems, VLM systems, computer vision, reinforcement learning, Claude Code, OpenAI Codex, Cursor, CoreML, MediaPipe

Quantitative research: Backtesting, systematic trading strategies, crack spread analysis, market data analysis, forecast-driven trading

Robotics and aerospace: ROS2, SLAM, path planning, PID control, Gazebo, MuJoCo, Isaac Sim, PX4, ArduPilot, Betaflight, Unreal Engine 5

Hardware and product: Arduino, VEX, Fusion 360, Onshape, 3D printing, CNC machining, woodworking, metal cutting, band saw, miter saw, SwiftUI, Supabase, Firebase, Stripe